

## LAB-3 : PROBLEM STATEMENT 14

### Healthcare & Telemedicine Corporate Mental Wellness & Counseling Platform

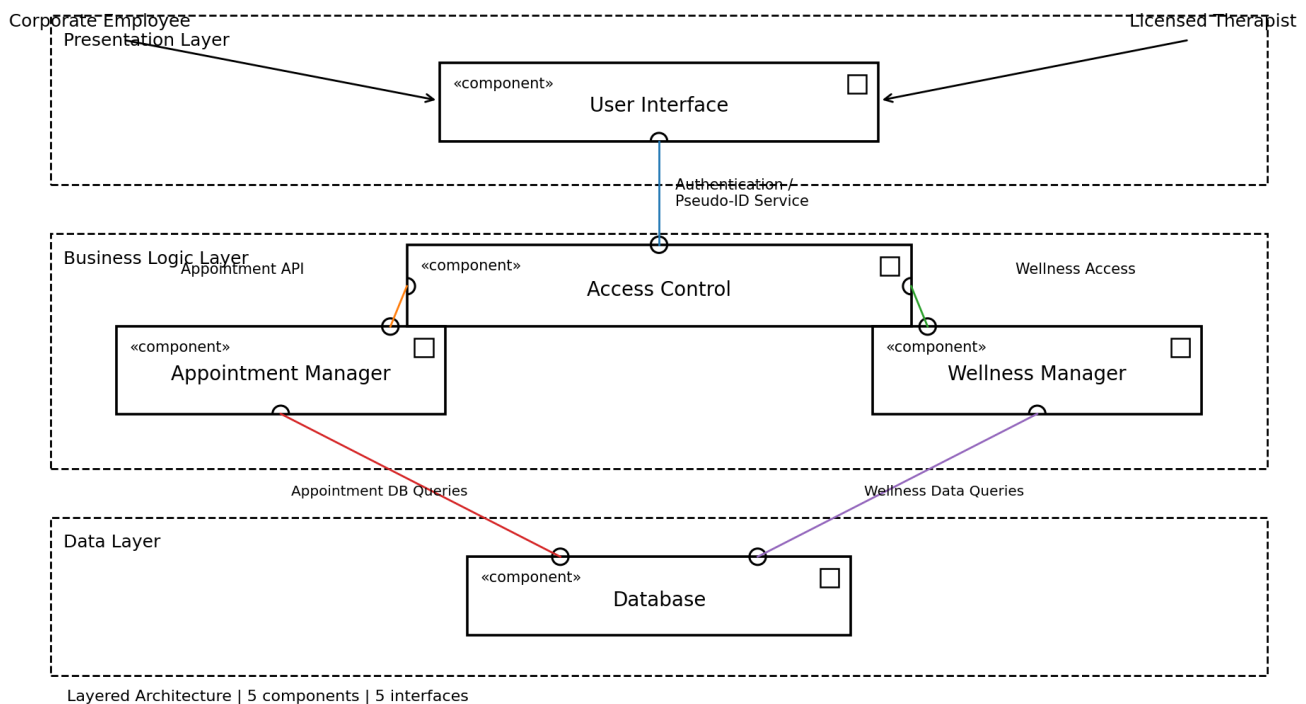
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## 1. UML COMPONENT DIAGRAM

Selected Architecture: Layered Architecture

The diagram contains the five main components and the required interfaces between them.



**Components:** User Interface, Access Control, Appointment Manager, Wellness Manager, Database.

**Interfaces:** Authentication / Pseudo-ID Service, Appointment API, Wellness Access, Appointment DB Queries, Wellness Data Queries.

## **2. WRITTEN ARCHITECTURAL JUSTIFICATION**

### **Architecture Selection**

I selected Layered Architecture because it separates the user interface, application logic, and database into different layers. It is simple to understand and suitable for this system.

#### **Reason 1 – Separation of responsibilities**

The interface, business logic, and database are separated, making the system organized and easier to manage.

#### **Reason 2 – Easier maintenance**

Each component has its own main task, so changes can be made to one part without changing the whole system.

#### **Security Advantage**

The Access Control component creates a pseudo-ID for every employee. Therapists only see the pseudo-ID, which keeps the employee identity confidential.

#### **Performance Benefit**

Appointments and wellness records are handled by separate components, so requests are processed in an organized way.

#### **Lab 3 Requirements Covered**

- 5 components included.
- 5 interfaces included.
- Provided and required interface notation shown.
- Component interactions and data flow shown.